SE

REFRACTIVE SURGERY

StandardEyes
Refractive surgery analysis

User manual & walkthrough · v1.0.0 · standardeyes.github.io

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What this app does

One tool from raw Excel to publication-ready tables and figures

1 **Compare groups**

Summary tables for any number of groups, with the right test chosen per variable type.

2 **Match fairly**

Matching with MatchIt (nearest neighbour on the propensity score, exact, coarsened exact or subclassification), with a live matched-count preview for nearest neighbour and balance reporting.

3 **Model properly**

Clustered GEE so both eyes of a patient are handled correctly, plus covariate-adjusted regression.

4 **Publish**

The StandardEyes figures — every group on one chart, each panel separately and combined, JCRS double-angle astigmatism plots, semicircle vector plots and stability over time.

Before you start

Almost everything is already inside the folder

You need R 4.5.x

The statistics run in R, so R must be installed on your computer.

Use R 4.5.x, not 4.6 or an older R. CRAN's main page now offers 4.6. R 4.5.2 is at cran.r-project.org/bin/windows/base/old/4.5.2/

Typical location:

C:\Program Files\R\R-4.5.2\

The app finds it automatically. If it can't, it asks you — see the next slide.

Already included

- The Python runtime that serves the app
- All 48 R packages the analysis needs
- The full statistical engine and figure code

Nothing is installed on your machine and nothing is downloaded — it works on a restricted network.

Starting the app

Copy once, then start it the same way every time

1

Copy the whole folder

Download StandardEyes-windows.zip from standardeyes.github.io, right-click it, choose Properties and tick Unblock. Extract the whole folder where you can write, e.g. C:\StandardEyes or Documents, not C:\Program Files.

2

Double-click the launcher

Start StandardEyes.bat

3

Work in your browser

It opens automatically on a free port, shown in the console window — so it never clashes with another app.

Keep the black console window open while you work. Closing it stops the app.

If the app can't find R

A one-time Setup screen appears — you can also reopen it any time

You have three ways to point it at R:

- **Pick from the list** — Any R installation found on the computer is listed — click one.
- **Browse…** — Opens a file picker. Choose Rscript.exe inside R's bin folder.
- **Paste the path** — Type it in and press Use this.

The app checks the choice immediately and reports the R version it found, with a warning if it is not 4.5.x.

No R at all? Install R 4.5.2 for Windows from cran.r-project.org/bin/windows/base/old/4.5.2/ (the main CRAN page offers a newer R), then click Re-check. No extra R packages are ever needed — this app brings its own.

Reopen later with the R setup button at the top right.

The workflow

Eight tabs, left to right — top to bottom here

1

Data & Groups

Load Excel, define groups

2

Variables & Types

Sections, types, VA notation

3

Filters & Subsets

Inclusion rules + subsets

4

Matching

Propensity-score matching

5

Regression

Adjusted GEE

6

GEE

Clustered comparison

7

Special outputs

Column roles, figures, export tabs

8

Run & Results

Run, preview, save

Steps 1–2 · Data and variables

1 Data & Groups

Load your Excel file (.xlsx or .xlsm) with Browse & load Excel in the left sidebar, then pick the sheet. The file stays on your computer.

Choose the column that defines the groups, then map its values into the groups you want to compare.

Set the reference group. Matching, GEE, regression and the figures' versus-reference lines compare each group with it.

Also set the patient ID, so both eyes of one patient are clustered correctly.

2 Variables & Types

Put each variable into Baseline, Preop / Intraop or Outcome, and give it a type:

continuous · ordinal · dichotomous · categorical

The type decides which test is used and how the value is displayed, so this is the step that shapes your tables.

Also say how your file writes visual acuity: decimal, logMAR, Snellen or ETDRS letters. CF, HM, LP and NLP become missing. If most of a column cannot be read in that notation, the run stops and says why.

Steps 3–4 · Filters and matching

3 Filters & Subsets

Add inclusion rules — age, follow-up, first treatment, cylinder limits, and so on. A blank cell passes a rule (add a notna rule to drop blanks), and rules use the values as recorded in your file.

Row counts update live, per group, so you can see exactly how many eyes each rule removes before running anything.

Each enabled subset (e.g. PRK and LASIK) gets its own full analysis. Rows outside every subset are left out. The Workbooks setting can add, or give only, one pooled workbook.

4 Matching

Matching against the reference group: nearest neighbour, exact, coarsened exact or subclassification.

Choose the covariates to match on and, for nearest neighbour, the ratio and the caliper (default 1:1, caliper 0.2 SD of the propensity score). For nearest neighbour a live preview shows how many eyes match before you run. The other three methods are not previewed.

Balance is reported on each pairing's PSM sheet and in its love plot, so you can show the groups became comparable.

Steps 5–6 · Regression and GEE

How to handle groups that still differ at baseline

GEE is how the regression is fitted, not an alternative to it. The model is outcome ~ group + covariates; GEE additionally accounts for both eyes coming from the same patient.

- **Adjust for what is unbalanced**

  Put the baseline variables that still differ into the regression covariates. If the group effect survives, it is not explained by those confounders.

- **Match first, then adjust**

  Set Run on to Matched cohort (or Both) on the Regression tab to adjust the outcome model for whatever matching left imbalanced.

- **Choose covariates by balance, not by p**

  Add a covariate because it is still imbalanced (SMD above ~0.1), never because it changes the result.

- **Small groups carry a caution**

  A clustered model needs at least 6 eyes and 3 patients, otherwise it reports Not estimable. A group with 2-9 patients gets a caution under Check before using the results.

- **Pick the GEE cohort**

  On the GEE tab, run GEE on the matched cohort (the default), the unmatched one or both. Main GEE is the group effect alone. Adjusted GEE adds the covariates you list there.

Step 7 · Special outputs and figures

Map the Column roles at the bottom of the tab first. Nothing is guessed from column names.

The composite	EVERY group on one chart, readable up to about five. One overall test plus, with three or more groups, each group against the reference (Holm-adjusted), clustered by patient when a patient ID is set. Panel sets: Full (ten panels), Refractive (six), Cataract (four). Stability joins the Full set as an eleventh panel.
Double-angle plots	The JCRS-2025 astigmatism standard: centroid, total SD, 95% nonparametric convex polygon and a dashed 95% CI of the centroid, per group.
Semicircle plots	Single-angle polar vector diagrams — TIA, SIA, difference vector and correction index — groups superimposed on a grey background, kept for familiarity.
Export tabs	Optional worksheets in mEYEstro and AstigMATIC format, for verifying numbers against those tools.

Every panel is also saved as its own TIFF (300 dpi by default), and the composites are previewed on the Run & Results tab.

Step 8 · Run, preview, save

Preview first, or save at once

Set the output folder in the left sidebar. ▶ RUN & SAVE writes the results straight there.

Tick Preview only first to keep them inside the app. A temporary copy waits in the app's own folder until your next run or until you close the app.

Then press Save ALL to output folder. It also copies the figures, love plots and filtered CSV. Save this one copies a single workbook.

Progress and a full log are shown while R runs. Cancel stops it.

What you get

- Excel workbooks — summary, Publication Table, matching, GEE, regression, with a Contents and a Manifest sheet
- A matched-only workbook with every pairing
- A love plot per matched pairing: LovePlot_<prefix>_<subset>_<pairing>_<time stamp>.png
- A figure folder per run. The combined figures are previewed in the app
- The filtered dataset as CSV (optional)

When the regression runs, adjusted effects and p-values are embedded in the main outcome table.

Presets — one per study

Save every setting of a project and reload it in one click

A preset stores everything: groups, variable typing, filters, subsets, matching, regression and figure choices.

- **Save** — Name the study and click save. It is stored inside the app on your own computer.
- **Reload** — Load the Excel file, then click the preset to restore the entire configuration. A preset never loads the data file itself.
- **Share** — Click Download config JSON and send it to a colleague. They click Load config JSON and get your exact setup. The data file is not inside, so they load their own copy of it.

Seven worked ‘Example – …’ presets are included. Load Example Data\Example_Refractive_Dataset.xlsx, click one, tick Preview only and press RUN. Presets you save appear beside them.

If something goes wrong

“Windows protected your PC”	Click More info, then Run anyway. To avoid it, tick Unblock in the ZIP’s Properties first.
“Windows security blocked a component”	Smart App Control or an antivirus blocked the runtime. Ask IT to allow this folder.
“Python runtime is missing”	The folder was copied incompletely, or is still inside a ZIP. Copy the whole folder again.
“Cannot write to its own folder”	Move the whole folder to Documents or C:\StandardEyes, not C:\Program Files.
The browser shows nothing	Wait a few seconds, then reload the address printed in the console window.
“R is not set up yet”	Click ⚙ R setup at the top right and point it at Rscript.exe.
A red “out of date” bar	Close the console window, start the app again, then press Ctrl+F5.
The analysis fails immediately	Check that R is version 4.5.x and the VA notation of your file. The log says why.
Am I in the right app?	The header shows StandardEyes and a version such as v1.0.0 (build 2026-10-02-se.9).

Everything runs on your own computer.

No data leaves the machine, no internet connection is needed, and no packages are ever installed.

Newer versions: standardeyes.github.io

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